BID: Secure Blockchain Based Identity Management System

Software Project Lab 3 Technical Report

Submitted by

Swarna Islam

BSSE-1214

Institute of Information Technology

University of Dhaka

Supervised by

Dr. Sumon Ahmed

Associate Professor

Institute of Information Technology

University of Dhaka

Submission Date: 01 December 2024



Letter of Transmittal

01 December, 2024

BSSE 4th Year Exam Committee

Institute of Information Technology

University of Dhaka

Subject: Technical Report Submission - BID: Secure Blockchain Based Identity Management System

Sir,

I am submitting the technical report for our Software Project Lab 3, "BID: Secure Blockchain Based Identity Management System."

This report comprehensively covers the project's technical details, including design, implementation, and testing phases. While I've tried to do my best, I welcome your feedback for improvement.

Thank you for your consideration.

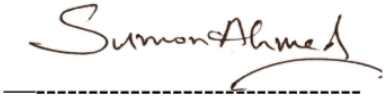
Sincerely,

Swarna Islam

BSSE-1214

Institute of Information

University of Dhaka



Supervisor's Signature

Acknowledgement

I am grateful to the Institute of Information Technology for giving me the opportunity to work on the project "BID: Secure Blockchain Based Identity Management System".

I would like to convey my tremendous gratitude to my supervisor, Dr.Sumon Ahmed, Associate Professor, Institute of Information Technology, University of Dhaka, for providing me with guidance and support throughout the project. His expertise and insights were instrumental in shaping the direction of the project. Lastly, I would like to thank my classmates. They have always been helpful and provided valuable insights from time to time.

Abstract

Traditional identity management systems face several challenges, including the risk of identity theft, dependency on centralized storage, and the need for physical identity cards to access services. Additionally, users often encounter limitations in accessing services outside their issued area, and compromised identity cards can be misused to retrieve sensitive information.

Block-NID is a secure blockchain-based identity management system designed to address these shortcomings through decentralization, user-centric control, and cryptographic security. Users can easily apply for a new identity through an intuitive user interface. Their personal data is encrypted using homomorphic encryption, ensuring that it remains secure even during processing. The encrypted data is stored in IPFS (InterPlanetary File System), and a request to add a new block containing encrypted biometrics, name, date of birth, and parent identities is sent to the blockchain network.

In this consortium blockchain, miner nodes randomly select requests from the mempool, verify the information, and broadcast validated blocks across the network. The system enables users to securely access services by verifying their identities through service nodes. These nodes validate the user's identity while ensuring that all transactions are encrypted and transparently recorded on the blockchain. Users are always aware of how their information is being accessed or utilized.

With its decentralized structure, transparent operations, and robust security measures, **Block-NID** redefines identity management, ensuring accessibility, privacy, and trust in the digital era.

BID: Secure Blockchain Based Identity Management System....0
Letter of Transmittal....1
Acknowledgement....2
Abstract....3
1. Introduction....6
1.1 Motivation....6
1.2 Scope....7
1.3 Project Timeline....7
1.4 Purpose of Document....8
2. Project Description....8
2.1 Quality Function Deployment....8
2.1.1. Normal Requirements....8
2.1.2. Expected Requirements....8
2.1.3. Exciting Requirements....9
2.2 Usage Scenario....9
3. Scenario Based Modeling....10
3.1 Use Case Diagram....10
3.1.1 BID....11
3.1.2 Modules of BID....12
3.1.2.1 PII Submission....12
3.1.2.2 Getting Block Identity....12
3.1.2.3 User Verification....13
3.1.2.4 Transaction Log....13
4. Clase Based Modeling....13
4.1 Analysis Class....13
4.2 Class Cards....14
CRC Card for KeyManager....14
CRC Card for UserManager....14
CRC Card for PIIProcessor....15
CRC Card for IdentityManager....15
CRC Card for PIIRetriever....15
CRC Card for MatchVerifier....16
4.3 CRC Diagram....17
5. Architectural Design....17
5.1 Architectural Context Diagram....18
5.2 Archetypes....19
5.3 Top Level Components....20
6. Preliminary Test Plan....20
6.1 High Level Description of Testing Goals....20

6.2 Test Cases..... 21
Conclusion..... 23
References..... 24

List Of Figures

Figure-1: Timeline for BID 7

Figure-2: UseCase Level 0 (BID) 11

Figure-3: UseCase Level 1 (Modules of BID) 12

Figure-4: CRC Diagram of BID 17

Figure-5: Architectural Context Diagram 18

Figure-6: Archetypes 19

Figure-7: Top Level Components 20

1. Introduction

In the realm of modern identity management, traditional systems have long served as a cornerstone for accessing essential services by providing individuals with unique identifiers. However, these systems face significant challenges, including centralized data storage, the need for physical identity cards, and vulnerability to identity theft. A critical concern arises from the risk of misuse if a card is lost, as embedded information can be extracted and exploited. Moreover, users often struggle to access services outside their issued area, further limiting the system's utility. Additionally, in traditional systems, users lack visibility into how their sensitive data is accessed, shared, or utilized, raising concerns over privacy and misuse.

Block-NID steps in as an innovative solution to these challenges, leveraging the power of blockchain technology to revolutionize identity management. Designed as a consortium blockchain, Block-NID decentralizes identity data storage, ensuring no single point of failure while enhancing transparency and security. Through an intuitive user interface, individuals can register for a new identity, with their data encrypted using homomorphic encryption.

Miner nodes validate user information and add encrypted identity blocks to the blockchain, enabling secure and verifiable transactions. Service nodes authenticate user identities, recording every transaction on the chain to ensure transparency and user awareness. Block-NID's model not only addresses the limitations of traditional systems but also empowers users with control and insight into how their sensitive data is being managed, fostering a secure, user-centric approach to identity management.

1.1 Motivation

The motivation behind **Block-NID** stems from the growing challenges and vulnerabilities in traditional identity management systems. Issues such as identity theft, misuse of lost identity cards, dependence on physical documents, and centralized data storage have raised significant concerns about privacy, accessibility, and security. Moreover, the lack of transparency in how sensitive user data is handled further exacerbates these problems, leaving users unaware of how their personal information is being accessed or utilized.

To address these challenges, **Block-NID** draws inspiration from the research conducted by F. Toutara and G. Spathoulas [1] and M. A. R. Tonu et al. [2]. The former explores a distributed biometric authentication system using blockchain and homomorphic encryption to enhance privacy and security. The latter introduces a blockchain-based conceptual model for national identity management, emphasizing decentralization and the use of IPFS for secure data storage.

Building on these foundational studies, **Block-NID** combines blockchain's decentralized nature with robust encryption techniques to ensure that users maintain control over their data. By

employing homomorphic encryption, the system guarantees that sensitive information remains secure even during processing. Inspired by these papers, Block-NID integrates encrypted biometric data, consortium blockchain, and transparent transaction recording to mitigate the risks associated with centralized identity management systems.

In essence, **Block-NID** is motivated by the urgent need for a secure, decentralized, and user-centric identity management system that addresses the limitations of traditional models. It aims to empower individuals with greater control over their identities, ensuring privacy, transparency, and accessibility in the digital age.

1.2 Scope

The scope of **Block-NID** is given below:

- **Block-NID** is a blockchain-based identity management system.
- It addresses traditional system issues by:
    - **Decentralization:** Eliminating central storage to prevent misuse and single points of failure.
    - **Transparency:** Recording all identity transactions on the blockchain for user visibility.
    - **Efficiency:** Reducing time, manpower, and costs in identity verification.
    - **Security:** Using homomorphic encryption and IPFS to safeguard sensitive data.
    - **Accessibility:** Allowing seamless service access without physical cards, regardless of location.

Block-NID ensures a secure, efficient, and user-focused identity management system.

1.3 Project Timeline

The estimated timeline of BIDis given below:

		Month 1				Month 2				Month 3				Month 4			
Tasks	In Weeks	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11	Week 12	Week 13	Week 14	Week 15	Week 16
Prerequisite Knowledge	2																
Requirements Specification	3																
Project Design	4																
Project Development	11																
Project Testing	8																
Final Report Writing	3																

Figure-1: Timeline for BID

1.4 Purpose of Document

The purposes of this document are-

- Identify and analyze the requirements
- Design the architecture
- Design the test plan
- Describe methodology
- Reduce the development effort
- Improve understanding

2. Project Description

The following depicts the overview of **Block-NID**'s Quality Function Deployment (QFD). The journey towards defining its QFD began with an in-depth review of relevant research literature, notably the works of F. Toutara and G. Spathoulas [1] and M. A. R. Tonu et al. [2], which provided valuable insights into secure blockchain-based identity management. This was further refined through analysis of traditional identity system challenges, such as inefficiency, high costs, identity theft, and lack of transparency, ensuring that **Block-NID** effectively meets user and system requirements.

2.1 Quality Function Deployment

Quality Function Deployment (QFD) is a technique that translates the needs of the customer

into technical requirements for the software. With respect to this project, the following requirements are identified-

2.1.1. Normal Requirements

1. Users should be able to register and authenticate in the system seamlessly.
2. Users should have control over what personal data is being processed.
3. Users should be able to access their stored information as needed.
4. Service providers should be able to access the relevant data for verification purposes.

2.1.2. Expected Requirements

1. Sensitive information, such as biometric data, should always be encrypted using FHE.
2. The system should have no single point of failure, ensuring decentralized and secure operations.

3. The application should be intuitive and user-friendly, ensuring easy navigation and usability.

2.1.3. Exciting Requirements

1. Transactions between users and service providers should be transparent, secure, and trackable.
2. Users could receive real-time notifications about access to their data by third parties.

2.2 Usage Scenario

The blockchain-based national identity system enables secure registration, privacy-preserving verification, and decentralized management of user identity data. Below is the usage scenario:

Registration

A user initiates the registration process by connecting to the identity management platform through their blockchain wallet. After establishing a connection, the user provides Personally Identifiable Information (PII), including their name, mother/father's name, date of birth, address, photo, and biometric information (e.g., fingerprint or facial scan).

The user also generates an FHE public-private key pair on their device and submits only the public key along with the PII to the identity management authority. The authority verifies the validity of the submitted information off-chain by cross-referencing existing records and validating official documents.

Once the PII is validated, the authority encrypts the user's sensitive information using the user's FHE public key. The encrypted data is then stored on the InterPlanetary File System (IPFS), and the Content Identifier (CID) is generated.

A new block is created containing the CID of the encrypted data, along with metadata such as the user's blockchain identity. The block is shared with the blockchain network for validation, and once 51% of the nodes reach a consensus, the block is added to the chain. The user is notified of the successful registration through their application.

Verification by Service Providers

When a service provider requires verification of a user's identity, they initiate a request by asking the user to submit their raw data (e.g., fresh biometric scan). The user provides the requested data via a secure interface.

The service provider encrypts the raw data using the FHE public key provided by the user during registration. They then retrieve the encrypted PII stored on IPFS through the CID and perform FHE-based computations to compare the encrypted raw data with the stored encrypted data.

The matching result, still in encrypted form, is modified by adding a random value to ensure the user cannot manipulate the final result. The encrypted result is sent back to the user with a request for decryption.

User Decision and Final Result
The identity management authority verifies the service provider's request and logs it as a transaction on the blockchain. The user receives a notification about the request and can either accept or reject it.

If the user accepts, they decrypt the matching result using their FHE private key on their device. The decrypted result is then encrypted using the service provider's public key to ensure that only the service provider can view it. The user sends the result back to the service provider, completing the verification process.

3. Scenario Based Modeling

Scenario-based modeling is an approach used in software development and design to create

models that depict how a system behaves or reacts in various real-world situations or scenarios.

These scenarios are typically based on user interactions, events, or conditions and help developers understand and visualize how the software will function in different contexts. By

simulating these scenarios, developers can better identify potential issues, make informed

design decisions, and ensure that the software meets user requirements across different usage

contexts.

3.1 Use Case Diagram

A use case diagram is a visual representation in software engineering that illustrates how a system interacts with its external entities or actors to achieve specific goals or tasks. It shows

the relationships between use cases (representing system functionalities) and actors (representing external users or systems). Use case diagrams are typically created through requirements gathering, where system functionalities and user interactions are identified, and then diagrammed using specialized software modeling tools.

3.1.1 BID

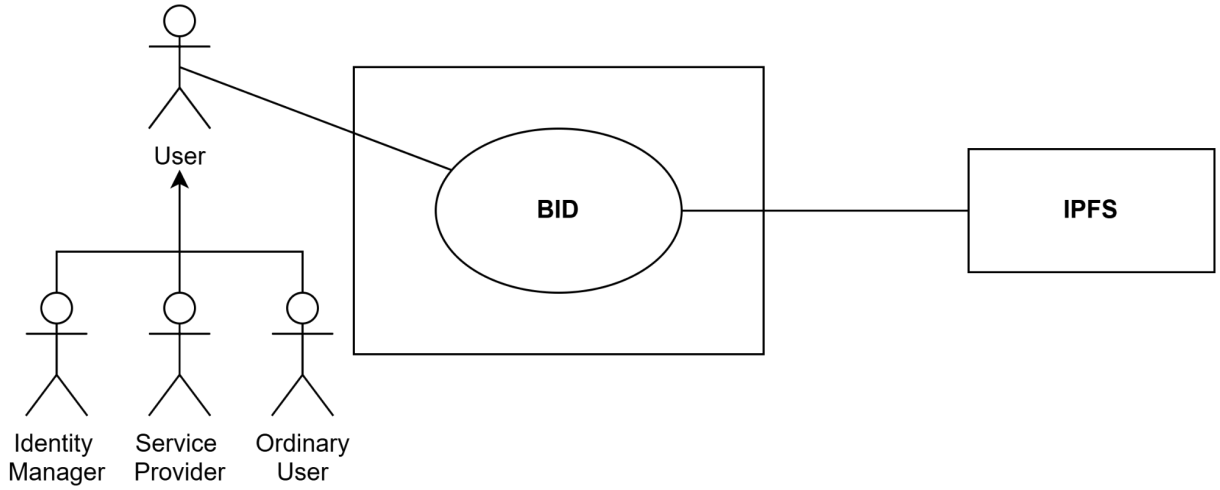


Figure-2: Use Case Level 0(BID)

Primary Actors:

- **User**: The individual registering their identity or interacting with the system for identity verification.

Secondary Actors:

- **IPFS:** PII Storage

Goal in Context:

The diagram shown above represents the **Blockchain-Based National Identity System**, where users securely register by submitting PII and their FHE public key. The system ensures encrypted data storage on IPFS, blockchain-based record management, and privacy-preserving identity verification using FHE computations.

3.1.2 Modules of BID

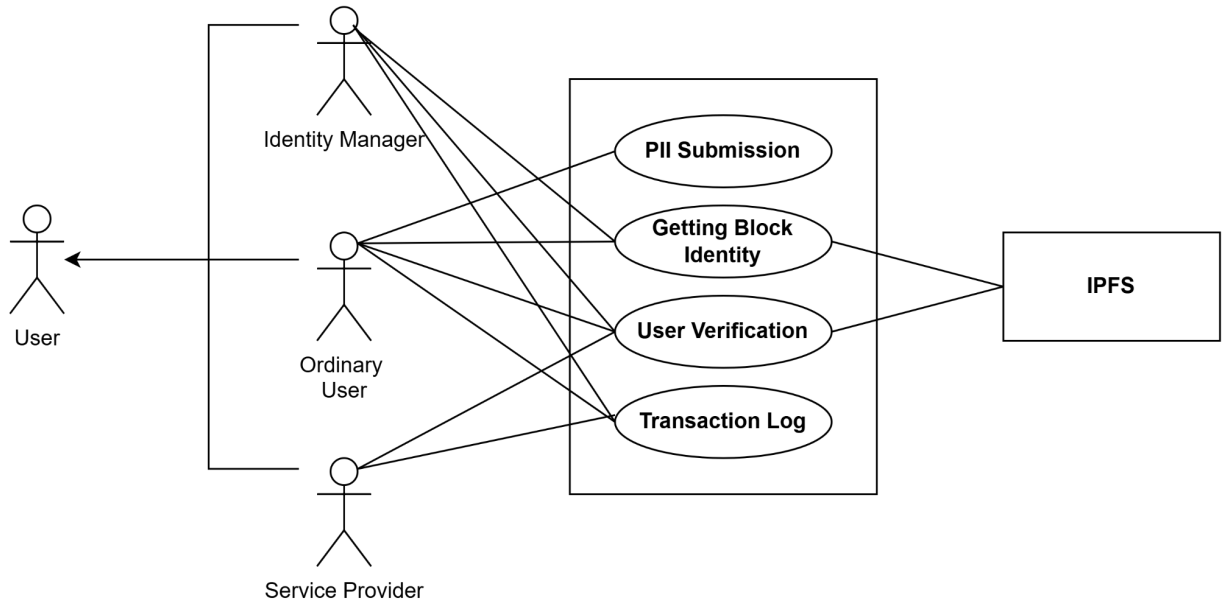


Figure-3: Use Case Level 1 (Modules of BID)

Primary Actor: Ordinary User, Identity Manager, Service Provider
Secondary Actor: IPFS

Goal in Context: The diagram shows the modules of a secure blockchain-based identity management system.

3.1.2.1 PII Submission

This module manages the initial registration process for the user. FHE(Fully Homomorphic Encryption) public and private key pair will be generated and stored in the user's device with password protection. In this module, it collects the necessary personally identifiable information (PII), biometric data (such as fingerprints and images), and the FHE public key. It validates the user's data off chain, ensuring it's in the correct format before submission to the identity management for encryption and storage.

3.1.2.2 Getting Block Identity

In this module, the submitted user data is encrypted using the FHE public key provided by the user. Once encrypted, the data is securely stored on IPFS, and an IPFS CID (Content Identifier) is generated. The encrypted data location is then added to a blockchain corresponding to the user's public key, ensuring a tamper-proof record of the user's encrypted data.

3.1.2.3 User Verification

This module facilitates the process where a service provider needs to verify a user's identity. The provider submits new encrypted biometric data for matching purposes. The system performs matching calculations using FHE, ensuring that the raw data is never compromised. This generates the encrypted result of the matching operation. The result is sent to the user with a random number added for security. This ensures that the user has control over the decryption and prevents tampering with the result before it is sent back to the service provider. The user can decrypt the matching result using their FHE private key. Once decrypted, the user can choose to accept or reject the result. If the user accepts, they encrypt the response with the service provider's public key before sending it back, ensuring the confidentiality of the decrypted result.

3.1.2.4 Transaction Log

This module ensures all actions taken during the registration, verification, and response processes are recorded. Every step, including encrypted data storage, verification requests, and user responses, is logged on the blockchain for transparency, auditability, and accountability.

4. Clase Based Modeling

Class-based modeling is a visual representation in software engineering that illustrates the structure of a software system using classes, their attributes, and relationships. It is commonly used in object-oriented design and modeling.

4.1 Analysis Class

After identifying nouns from the scenario, I have filtered nouns belonging to the solution domain using General Classification (External entities, Things, Events, Roles, Organizational units, Places, and Structures). Nouns selected as a potential class were filtered using Selection Criteria (Retained information, Needed services, Multiple attributes, Common attributes, Common operations, and Essential requirements). After performing analysis on potential classes, I have found the following analysis classes-

- User Manager
- PII Processor
- Identity Manager
- PII Retriever
- Match Verifier
- Key Manager

4.2 Class Cards

CRC Card for KeyManager

KeyManager	
Attributes	**Methods**
- password - user_public_address	+ generate_fhe_key() + store_keys()
Responsibilities	Collaborators
• Generate FHE key pair for encryption and decryption. • Manage the storage of the user's public and private keys.	

CRC Card for UserManager

UserManager	
Attributes	**Methods**
- password - user_fhe_public_key - user_fhe_private_key - user_public_address - cid	+ view_pii() + request_for_service() + decrypt_matching_result()
Responsibilities	Collaborators
• Allow the user to view their submitted PII. • Handle requests made by the user to access or use services. • Decrypt encrypted matching results sent by the service provider.	IdentityManager, MatchVerifier, KeyManager

CRC Card for PIIProcessor

PIIProcessor	
Attributes	**Methods**
- pii - user_fhe_public_key - user_public_address	+ retrieve_pii()() + verify_pii() + encrypt_pii() + store_on_ipfs()
Responsibilities	Collaborators
• Handle submission, verification, encryption, and storage of PII during registration.	UserManager

CRC Card for IdentityManager

IdentityManager	
Attributes	**Methods**
- cid - user_public_address	+ add_user() + notify_user()
Responsibilities	Collaborators
• Add the user to the blockchain network by generating and broadcasting a block.	PIIProcessor, UserManager

CRC Card for PIIRetriever

PIIRetriever	
Attributes	**Methods**
- cid	+ retrieve_new_pii()

- user_public_address - user_fhe_public_key	+ encrypt_new_pii() + retrieve_ipfs_encrypted_pii()
Responsibilities	**Collaborators**
● Retrieve new PII, encrypt it, and fetch stored encrypted PII from IPFS.	IdentityManager, UserManager

CRC Card for MatchVerifier

MatchVerifier	
Attributes	**Methods**
- user_fhe_public_key - user_public_address	+ calculate_match() + get_random_number() + verify_encrypted_match()
Responsibilities	**Collaborators**
● Calculate and verify the match between new and stored PII. ● Generate random numbers for security and verify encrypted results.	PIIRetriver, IdentityManager, UserManager

4.3 CRC Diagram

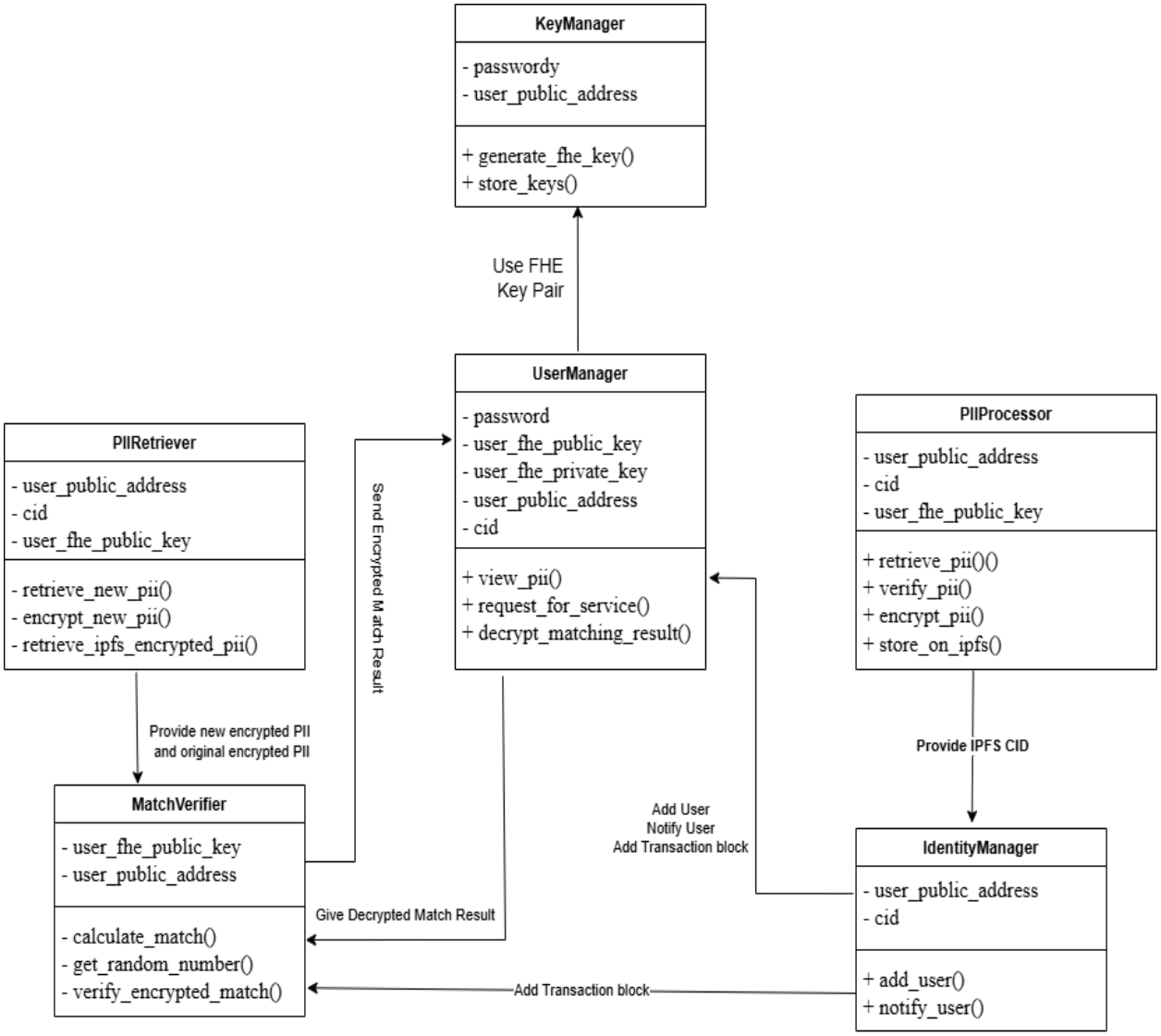


Figure-4: CRC Diagram of BID

5. Architectural Design

Architectural design is a visual representation in software engineering that outlines the high-level structure and organization of a software system. It focuses on defining the major components or modules of the system, their interactions, and the overall system's architecture.

5.1 Architectural Context Diagram

The architectural context diagram for the application is given below-

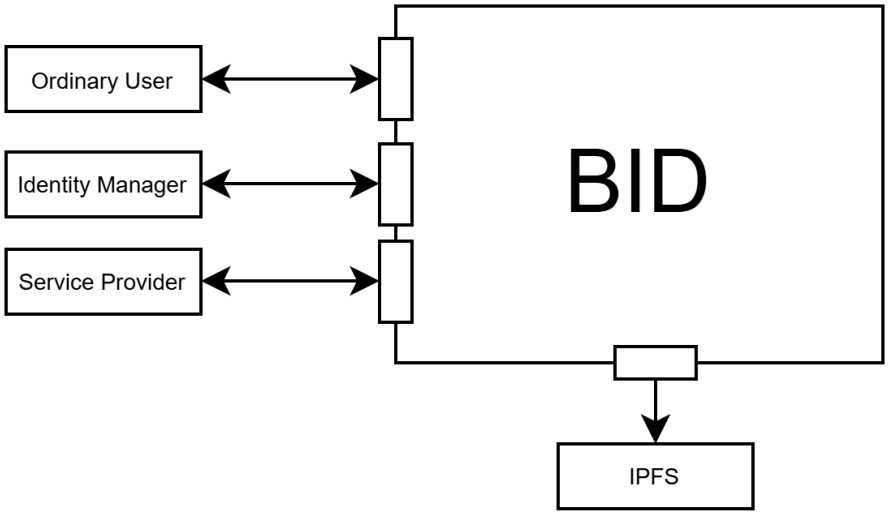


Figure-5: Architectural Context Diagram

The identity management system has three categories of users: **Ordinary Users**, **Identity Managers**, and **Service Providers**. Ordinary Users are the producers of information, submitting their personally identifiable information (PII) and biometric data. Identity Managers are responsible for validating, encrypting, and storing user data securely on IPFS and managing user registrations in the blockchain. Service Providers consume the information by verifying user identity for various services through encrypted matching processes.

The system interacts with one subordinate system, **IPFS**, which serves as the storage medium for encrypted user data. However, there are no superordinate or peer-level systems directly interacting with this system.

5.2 Archetypes

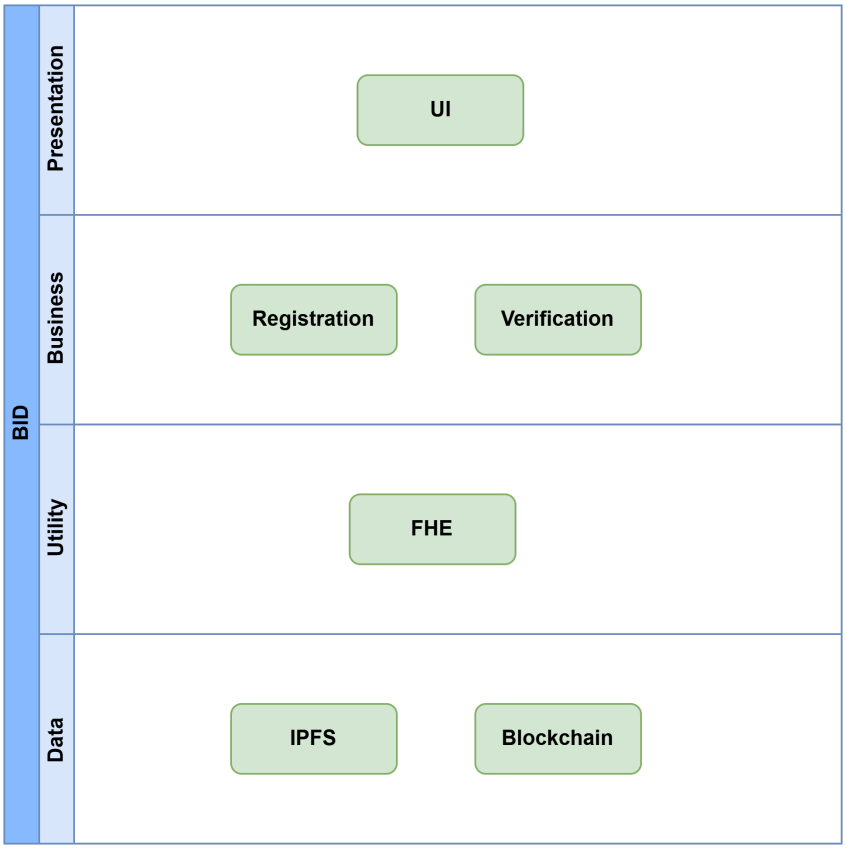


Figure-6: Archetypes

1. Presentation Layer

- **Purpose**: This layer handles the interaction between the user and the system.
- **Component (UI)**: It encapsulates the user interface for registration, identity verification, and result display.

2. Business Layer

- **Purpose**: Handles core business logic for registration and verification processes.
- **Components (Registration, Verification)**:
    - **Registration**: Manages the workflow of collecting and encrypting user data and storing it on IPFS.
    - **Verification**: Manages matching operations and ensures the verification results are accurate.

3. Utility Layer

- **Purpose**: Provides reusable services and frameworks that support the business layer (e.g., cryptographic operations).
- **Component (FHE)**: Handles Fully Homomorphic Encryption operations like key generation, encryption, decryption, and secure matching.

4. Data Layer

- **Purpose**: Handles data storage, retrieval, and logging operations.
- **Components (IPFS, Blockchain)**:
    - **IPFS**: Stores encrypted PII securely and provides content identifiers (CIDs).
    - **Blockchain**: Logs transactions for auditability and ensures tamper-proof data integrity.

5.3 Top Level Components

Overall architectural structure with top level components is illustrated below-

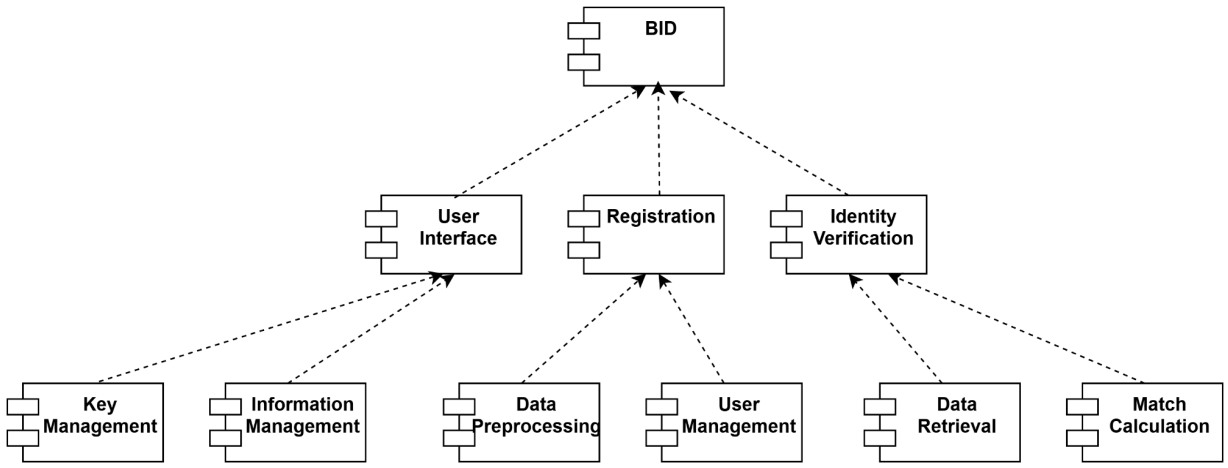


Figure-7: Top Level Components

6. Preliminary Test Plan

In this chapter, a high level description of testing goals and summary of features to be tested are presented.

6.1 High Level Description of Testing Goals

Testing goals for this system are represented below from a high level:

- To ensure that user data (PII and biometric information) is correctly collected, encrypted, and stored
- To ensure that Fully Homomorphic Encryption (FHE) operations, including key generation, encryption, decryption, and secure matching, work as expected
- To ensure that registration and verification processes are performed accurately and efficiently
- To validate that data stored on IPFS is retrievable using the Content Identifier (CID) and remains tamper-proof
- To confirm that all blockchain transactions, including registration and verification records, are logged accurately and securely
- To ensure that the UI presents clear, user-friendly prompts and feedback during all processes
- To ensure the system prevents unauthorized access and ensures data confidentiality and integrity
- To validate that matching results and verification responses are accurate and transmitted securely
- To ensure the system can handle edge cases, such as invalid data, interrupted transactions, or malicious attempts, gracefully
- To confirm that the system runs smoothly under expected and peak loads

6.2 Test Cases

Test Cases for the Application -

Test Case 1: Key Generation

Test Scenario:

- Generate FHE key pair on a new user's device during the registration process.

Expected Outcome:

- The system successfully generates a public-private FHE key pair.
- The keys are securely stored on the user's device with password protection.

Test Case 2: PII Submission

Test Scenario:

- Submit user PII and biometric data (e.g., fingerprint and photo) for registration.

Expected Outcome:

- The system validates and encrypts the PII data using the FHE public key.
- The encrypted PII is stored on IPFS, and a CID is generated without errors.

Test Case 3: User Registration

Test Scenario:

- Add the user to the blockchain network after successful PII submission.

Expected Outcome:

- A block containing the IPFS CID and user's public address is generated and broadcasted.
- After achieving 51% consensus, the block is added to the blockchain.

Test Case 4: User Verification

Test Scenario:

- A service provider submits a new encrypted PII for user identity verification.

Expected Outcome:

- The system retrieves encrypted PII from IPFS and performs matching using FHE without decrypting the data.
- The matching result is encrypted and sent to the user securely.

Test Case 5: Matching Result Decryption

Test Scenario:

- User receives the encrypted matching result and decrypts it with their private FHE key.

Expected Outcome:

- The matching result is decrypted accurately by the user.
- The user can either accept or reject the result and send it back to the service provider.

Test Case 6: Secure Data Storage

Test Scenario:

- Encrypt and store PII data on IPFS using the FHE public key.

Expected Outcome:

- The PII data is encrypted correctly.
- The CID generated from IPFS matches the location of the encrypted data.

Test Case 7: Blockchain Transparency

Test Scenario:

- Check if the system logs all operations (e.g., registration, verification requests, and responses) on the blockchain.

Expected Outcome:

- All transactions are accurately recorded on the blockchain.
- Transaction logs are tamper-proof and auditable.

Conclusion

In this report, I have discussed all the necessary technical details to develop the "Blockchain-Based National Identity System with FHE Encryption." I have covered the scope, assumptions, and usage scenarios of the system. Scenario-based modeling, data modeling, and class-based modeling have been performed to represent the structure and flow. The layered architectural design has been elaborated along with key technical components like FHE, IPFS, and blockchain. I have also presented a high-level description of testing goals and a summary of the features and functionalities to be tested. Figures and tables have been included to provide a clear visualization of the system's technical aspects.

Although this system effectively addresses the security, privacy, and reliability concerns in identity management, there is room for further improvements. Future scope includes introducing user information modification management, ensuring consistency and security in data updates without compromising privacy or system integrity.

With this document, I have aimed to present a comprehensive workflow of the entire system. I hope this report provides readers with a clear understanding of the technical details and practical implementation of the tool.

References

[1] M. A. R. Tonu, S. Hridoy, M. A. Ali and S. A. Azad, "Block - NID: A Conceptual Secure Blockchain Based National Identity Management System Model," *2019 IEEE Asia-Pacific Conference on Computer Science and Data Engineering (CSDE)*, Melbourne, VIC, Australia, 2019, pp. 1-7, doi: 10.1109/CSDE48274.2019.9162366.

[2] F. Toutara and G. Spathoulas, "A distributed biometric authentication scheme based on blockchain," *2020 IEEE International Conference on Blockchain (Blockchain)*, Rhodes, Greece, 2020, pp. 470-475, doi: 10.1109/Blockchain50366.2020.00068.